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Nuclear Waste Storage Sites in the United States

Congressional interest in nuclear waste is generally focused on managing commercial spent nuclear fuel (SNF), the waste produced from commercial nuclear power plants, and other high-level nuclear wastes (HLW), largely from Cold War–era nuclear weapons materials production. No country, including the United States, has a permanent geologic repository for disposal of commercial SNF and other HLW. Currently, commercial nuclear power plants generally store SNF on site, awaiting disposal in a permanent repository. The Nuclear Waste Policy Act of 1982 (NWPA; P.L. 97-425) authorized the Department of Energy (DOE) to site a geologic repository for the permanent disposal of high-level radioactive waste. Congress amended the NWPA in 1987 to designate Yucca Mountain, NV, as the only location to be considered by DOE to construct a national high-level nuclear waste repository. Political and legal opposition to the project has delayed the licensing, construction, and operation of the proposed Yucca Mountain repository. NWPA authorized DOE to enter into agreements with nuclear utilities and other reactor owners to collect fees to pay for DOE's disposal of the SNF. However, due to the delay in operation of a permanent repository, the federal government has paid $12.2 billion from the Judgment Fund to nuclear utilities and other reactor owners pursuant to court settlements and final judgments through September 30, 2025. **Figure 1** shows the locations of 80 sites in the United States where nuclear waste is currently stored. At 54 of these sites, 95 operating nuclear reactors generate approximately 20% of the total annual electricity production for the United States. A "site" in **Figure 1** is a geographically distinct location. In some instances, multiple nuclear reactors may be co-located (operating or not) at a single site.

Generation of Spent Nuclear Fuel and Other High-Level Nuclear Waste

Commercial civilian nuclear reactors use low-enriched uranium oxide to generate nuclear power through fission reactions. The heat from fission in the reactor drives a steam turbine and a generator. Over time, the reactor fuel becomes incapable of economically producing power (i.e., "spent") and must be replaced. Fuel assemblies removed from the reactor following power production are called *spent nuclear fuel*. SNF is stored in one of two ways: in wet storage pools and in dry casks. Immediately following power production, SNF is discharged from the reactors and stored in wet storage pools on site, as it remains intensely radioactive and thermally hot. Wet pools provide regulated conditions allowing the SNF to cool while water is circulated and maintained to keep it from boiling off and uncovering the SNF. Some sites store SNF using a combination of wet storage and dry casks. Following the initial cool-down period, most facilities transfer SNF to dry casks and store it on site. Dry cask storage of SNF does not require active cooling dependent upon pumps. Plant operators place dry casks within a security perimeter outside of the reactor building, either vertically on a concrete pad or within an engineered concrete bunker. HLW stored at DOE sites is characteristically different from SNF produced at commercial nuclear reactors. The historical production of nuclear warheads by DOE and its predecessor agencies generated most HLW, which consists of radioactive liquids from reprocessing SNF and extracting special nuclear materials. DOE manages most HLW as liquids in underground tanks. DOE has converted some HLW to a powder form and some HLW has been solidified in glass for future disposal.

Nuclear Waste Storage

Commercial nuclear power plants have been storing SNF on site pursuant to regulations promulgated by the Nuclear Regulatory Commission under the Atomic Energy Act, as amended (P.L. 83-703). The U.S. Nuclear Waste Technical Review Board estimates the quantity of SNF stored as of 2012 at commercial nuclear power plants will approximately double by 2048.

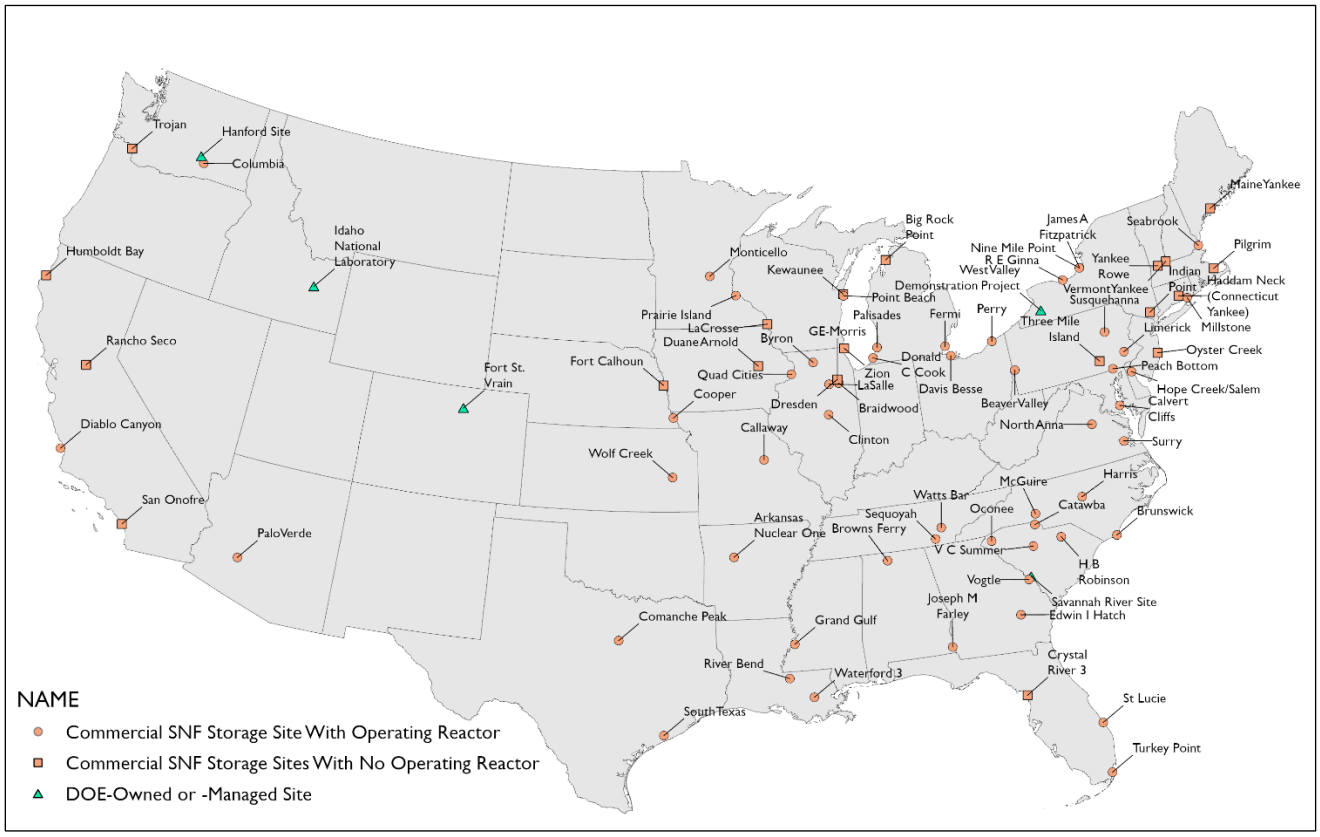
Stranded Sites

At 26 nuclear waste storage sites (33% of the total sites), there are no ongoing reactor operations. These "stranded sites" are facilities that store nuclear waste but lack an operating reactor generating power and revenue. *Stranded* does not imply that the waste has been abandoned or lacks regulatory oversight. Sites where operating and shutdown reactors are co-located are not considered stranded sites regardless of the presence of stored SNF from a co-located closed reactor.

Issues for Congress

Some Members of Congress have introduced legislation in the current and over the past several Congresses intended to identify potential solutions to the challenge of long-term management and storage of SNF. Those bills have varied widely in scope and effect and included authorizing DOE to initiate a program for developing consolidated interim nuclear waste storage sites, research and funding for reprocessing of SNF, and repealing the provision in the NWPA selecting the Yucca Mountain site, among others. The Trump Administration has supported alternative SNF storage proposals to Yucca Mountain. Subject to an executive order, DOE has entered Memoranda of Understanding with five states—Utah, Tennessee, Oklahoma, Louisiana, and Idaho—to establish *Nuclear Lifecycle Innovation Campuses.* According to DOE, the "proposed campuses could support activities across the full nuclear fuel lifecycle, including fuel fabrication, enrichment, reprocessing used nuclear fuel, and disposition of waste."

Figure 1. Nuclear Waste Storage Sites in the United States



Sources: Compiled by CRS using various U.S. Nuclear Regulatory Commission (NRC), Department of Energy (DOE), and other sources, including NRC, *Operating Nuclear Power Reactors (by Location or Name)*, updated August 27, 2026, https://www.nrc.gov/facilities-safety/facility-finder/reactors, *Evaluation of Options for Permanent Geologic Disposal of Spent Nuclear Fuel and High-Level Radioactive Waste in Support of a Comprehensive National Nuclear Fuel Cycle Strategy*, FCRD-UFD-2013-000371, Revision 1; SAND2014-0187P (vol. I); SAND2014-0189P (vol. II), April 15, 2014; DOE, *Report to Congress on the Demonstration of the Interim Storage of Spent Nuclear Fuel from Decommissioned Nuclear Power Reactor Sites*, DOE/RW-0596, December 2008; Frank Marcinowski, *Overview of DOE's Spent Nuclear Fuel and High-Level Waste: Presentation to the Blue Ribbon Commission on America's Nuclear Future*, DOE, March 25, 2010.

Notes: *Nuclear waste* refers to spent nuclear fuel from commercial nuclear power plants and other high-level nuclear waste. The locations of research reactor sites, special nuclear materials (e.g., plutonium-239 and uranium-235), transuranic wastes, or low-level nuclear wastes were not included in this analysis. Although currently listed as commercial spent nuclear fuel (SNF) storage sites without an operating reactor (often referred to as "stranded"), Three Mile Island, PA (known now as "Christopher M. Crane Clean Energy Center"), and Duane Arnold, IA, have formally requested to NRC to restart operations. No nuclear waste storage sites are located in Alaska or Hawaii.

For more information, see these other CRS products:

- CRS Report R48364, *Considerations for Reprocessing of Spent Nuclear Fuel*, by Lance N. Larson and Mark Holt.
- CRS Report R42853, *Nuclear Energy: Overview of Congressional Issues*, by Mark Holt.

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